

Name: _____

Class: _____

ACTIVITY SHEET

5.4 Working with moles

For all calculations, use $N_A = 6.02 \times 10^{23}$.

1 Calculate the number of moles of each of these substances.

a 3.01×10^{23} atoms of helium

b 12.04×10^{24} molecules of carbon dioxide

c 18.06×10^{22} formula units of copper(II) nitrate

2 Calculate the number of particles in each of these substances.

a 2.50 mol of carbon atoms

b 0.01 mol of sulfur dioxide molecules

c 12.5 mol of sulfate ions

- 3** A sample contains 4.0 mol of sulfuric acid (H_2SO_4).
- a** Find the number of moles of each type of atom present.

 - b** Calculate how many moles of atoms are present altogether.

 - c** Determine how many individual atoms are present.
- 4** A sample contains 0.5 mol of calcium sulfate (CaSO_4).
- a** Find the number of moles of each type of atom present.

 - b** Calculate how many moles of atoms are present altogether.

 - c** Determine how many individual atoms are present.

- www.nelsonnet.com.au